

ARACHIS ONLINE TRAINING

WHITE EMULSION PAINT MAKING MODULE (20 LITRES)

INTRODUCTION

Welcome to the Arachis Online Training Paint Manufacturing Module.

In this lesson, you are going to learn how to make ****White Emulsion Paint****, one of the most common paints used on walls and ceilings in homes, schools, offices, shops, and many buildings.

This module is written for complete beginners, so every ingredient and step is explained simply.

By the end of this lesson you should understand:

- * What each ingredient does
- * How to mix the paint properly
- * Common mistakes beginners make
- * How to improve quality
- * How to package and sell the paint

WHAT IS EMULSION PAINT?

Emulsion paint is a ****water-based paint****.

This means:

- * Water is used during production
- * It dries faster
- * It has less smell than oil paint
- * It is cheaper and easier to make

White emulsion paint is mainly used for:

- * Interior walls
- * Ceilings
- * Schools
- * Churches
- * Shops
- * Homes

SAFETY FIRST

Even though emulsion paint is safer than oil paint, you must still protect yourself.

Wear:

- * Gloves
- * Face mask
- * Apron or old clothes

Avoid:

- * Breathing powder dust
- * Drinking ingredients
- * Mixing with bare hands

Always work in a clean ventilated area.

EQUIPMENT NEEDED

Before starting, prepare the following:

Equipment	Purpose
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25L bucket	Mixing
Mixing stick or drill mixer	Stirring
Measuring containers	Accurate measurements
Sieve	Removing lumps
Scale	Measuring powders
Storage containers	Packaging

INGREDIENTS FOR 20 LITRES WHITE EMULSION PAINT

Ingredient	Quantity	Purpose
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Clean water	10 litres	Main liquid
Calcium Carbonate (Whiting)	8 kg	Main filler/body
Titanium Dioxide	1 kg	Gives whiteness and coverage
PVA Binder / Acrylic Binder	2 litres	Helps paint stick to walls
Thickener (CMC or HEC)	150–200g	Thickens the paint
Defoamer	30–50ml	Removes foam/bubbles
Preservative/Biocide	50ml	Prevents rotting
Texanol (optional)	100ml	Improves finish and durability

UNDERSTANDING THE INGREDIENTS

1. WATER

Water is the main liquid carrier.

Use:

- * Clean water
- * Not salty water
- * Not dirty water

Dirty water can spoil paint quality.

2. CALCIUM CARBONATE (WHITING)

This is the main body of the paint.

It:

- * Makes paint thick
- * Increases volume
- * Reduces production cost

Without it, paint becomes too watery.

3. TITANIUM DIOXIDE

This is the ingredient that makes paint:

- * Very white
- * Bright
- * Able to cover walls properly

This is one of the most expensive ingredients.

Less titanium dioxide = cheaper but poorer quality paint.

4. PVA / ACRYLIC BINDER

This is one of the MOST IMPORTANT ingredients.

The binder:

- * Makes paint stick to the wall
- * Prevents powdering
- * Improves durability

If binder is too little:

- * Paint washes away easily
- * Paint peels off
- * Paint becomes dusty

5. THICKENER (CMC OR HEC)

This controls thickness.

Too little:

- * Paint becomes watery

Too much:

- * Paint becomes too thick

Always add slowly.

6. DEFOAMER

When mixing paint, foam forms.

Defoamer:

- * Removes bubbles
- * Makes paint smooth

7. PRESERVATIVE

Paint contains water.

Water can allow:

- * Bacteria
- * Fungus
- * Bad smell

Preservative prevents this.

8. TEXANOL (OPTIONAL)

Texanol improves:

- * Smoothness
- * Film formation
- * Quality
- * Professional finish

Premium paints usually contain texanol.

STEP-BY-STEP PROCEDURE

STEP 1 — ADD WATER

Pour:

- * 10 litres of clean water

into a clean mixing bucket.

STEP 2 — ADD THICKENER

Slowly add:

- * 150–200g CMC or HEC

while stirring continuously.

IMPORTANT:

Do NOT pour all at once.

If added too fast:

- * Lumps form
- * Thickener fails to dissolve

After mixing:

Leave it for 30–60 minutes to swell properly.

This step is VERY IMPORTANT.

STEP 3 — ADD CALCIUM CARBONATE

Slowly add:

- * 8 kg calcium carbonate

while mixing continuously.

Do not dump everything at once.

Mix until smooth.

No lumps should remain.

STEP 4 — ADD TITANIUM DIOXIDE

Add:

- * 1 kg titanium dioxide

Mix thoroughly.

This stage improves:

- * Whiteness
- * Opacity
- * Covering power

STEP 5 — ADD BINDER

Add:

- * 2 litres PVA or acrylic binder

Mix thoroughly.

This stage gives the paint:

- * Adhesion
- * Strength
- * Durability

STEP 6 — ADD OTHER CHEMICALS

Now add:

- * Defoamer
- * Preservative
- * Texanol

Mix properly.

STEP 7 — CHECK THICKNESS

Observe the paint carefully.

If too thick:

- * Add small water slowly

If too thin:

- * Add small thickener solution

Do NOT overcorrect quickly.

Adjust slowly.

STEP 8 — SIEVE THE PAINT

Pass the paint through a sieve.

This removes:

- * Lumps
- * Dirt
- * Undissolved particles

This improves final quality.

STEP 9 — PACKAGE

Package into:

- * 1L
- * 2.5L
- * 5L

- * 20L containers

Seal tightly.

COMMON BEGINNER MISTAKES

1. Adding Thickener Too Fast

Result:

- * Lumps

Solution:

- * Add slowly while stirring

2. Using Too Much Water

Result:

- * Weak paint
- * Poor coverage

3. Using Too Little Binder

Result:

- * Paint peels
- * Paint powders

4. Poor Mixing

Result:

- * Uneven paint
- * Lumps

5. Using Dirty Containers

Result:

- * Spoiled paint
- * Bad smell

Always use clean equipment.

HOW TO TEST YOUR PAINT

Apply small paint on:

- * Board
- * Wall
- * Cardboard

Check:

- * Whiteness
- * Smoothness
- * Coverage
- * Drying speed

HOW TO MAKE PREMIUM QUALITY PAINT

To improve quality:

Ingredient	Improvement
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Increase binder	Better durability
Increase titanium dioxide	Better whiteness
Use acrylic binder	Premium finish
Add texanol	Smoother film

LOW-COST FORMULA STRATEGY

If targeting low-income customers:

You can reduce:

- * Titanium dioxide from 1kg to 500g

But remember:

- * Coverage reduces
- * More coats needed

Never reduce binder too much.

Cheap paint that peels destroys customer trust.

STORAGE

Store paint:

- * In closed containers
- * Away from sunlight
- * In cool areas

Good paint can last several months if preserved properly.

BUSINESS TIPS

You can sell paint to:

- * Builders
- * Schools
- * Churches
- * Shops
- * Hardware stores
- * House owners

Small buckets move faster in low-income areas.

Example:

- * 1L
- * 2.5L

can sell quickly.

FINAL ADVICE FROM ARACHIS ONLINE TRAINING

Do not rush production.

Good paint comes from:

- * Proper measurements
- * Proper mixing
- * Patience

* Clean equipment

Start small.

Practice repeatedly.

Improve gradually.

Consistency builds a strong paint brand.

END OF MODULE

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